B2 — A Recommender from Scratch on MovieLens

Proposal

Task. Implement BPR matrix factorization (MF + BPR) and a LightGCN graph recommender (LightGCN + BPR) from scratch. Test whether graph propagation improves top-K recommendation and how performance changes when explicit low ratings replace a proportion of unrated training negatives.

Dataset. Use the fixed MovieLens-1M release: 1,000,209 timestamped ratings from 6,040 users on 3,706 rated movies (dataset reference). Download directly from GroupLens; do not redistribute. Primary label policy: **4–5 = positive**, **1–2 = explicit negative**, and **3 = neutral**.

Methods and controlled experiment. Build the user–item bipartite graph from training positives only. MF + BPR uses dot-product user/item embeddings; LightGCN + BPR propagates embeddings over the normalized graph and combines layer representations. Train both with BPR loss, comparable embedding dimensions, regularization search spaces, training budgets, and hyperparameter-tuning budgets.

Include a most-popular baseline based on positive interaction counts in train. Compare two models under three target proportions of explicit negatives:

Model	A: 0% explicit negatives	B: 25% explicit negatives	C: 50% explicit negatives
MF + BPR	100% unrated	75% unrated + 25% training ratings of 1–2	50% unrated + 50% training ratings of 1–2
LightGCN + BPR	100% unrated	75% unrated + 25% training ratings of 1–2	50% unrated + 50% training ratings of 1–2

Complete the four configurations with 0% and 50% first, then add 25% for both models as a sensitivity experiment. The full planned comparison therefore contains **six configurations plus Popularity**.

Negative sampling. Use the same training-defined movie catalog across configurations. For each user:

- The explicit-negative pool contains catalog movies rated 1–2 by that user in train.
- The unrated pool contains catalog movies with no rating from that user in train.
- All training-rated movies, including ratings of 3, are excluded from the unrated pool.
- Validation and test interactions do not affect either training pool.

For a target explicit-negative proportion (p), select the explicit pool with probability (p) and the unrated pool otherwise, then sample uniformly within the selected pool. If the user has no eligible explicit negatives, fall back to unrated-only sampling. Repeated sampling of the same negative across training steps is allowed.

Keep the number of training triplets and optimizer updates comparable across configurations. Report the realized proportion of explicit negatives, which may be lower than the target because

of fallback. Future positive interactions may be sampled as unrated negatives; acknowledge this limitation rather than using future information to exclude them.

Temporal split. Use two common timestamp boundaries for all users: train before the first boundary, validation between the boundaries, and test after the second boundary. Apply the split to **all ratings**, then derive positives, explicit negatives, and neutral observations within each period. Keep records with identical timestamps in the same partition.

Choose and document the boundaries after a coverage audit of period sizes, eligible users, and positives per user. Freeze the split before model tuning; do not choose boundaries based on model performance.

Evaluate all eligible positive interactions in each held-out period. Multiple positives per user are desirable, but users with only one eligible positive remain included. The previous leave-two-positive-out requirement of at least three positives does not apply.

Catalog and evaluation eligibility. Define the movie catalog as movies with at least one positive interaction in train. Use this same catalog for all configurations and Popularity.

A user is eligible for evaluation in a given period if they have at least one training positive and at least one positive target in that period remaining in the candidate set. Exclude targets outside the catalog from the primary metrics. Report excluded users and interactions, including those excluded because they lack training history or eligible targets. The primary evaluation therefore concerns users and movies with positive training history.

Masking and evaluation. Exclude all movies previously rated by the user, regardless of rating:

- At validation, mask all movies rated by that user in train.
- At test, mask all movies rated by that user in train or validation.

Generate one ranking per user at the start of each evaluation period and evaluate it against all eligible positives in that period. Do not use interactions from within the evaluated period to update rankings or masks.

Select hyperparameters and checkpoints using validation NDCG@20. Do not retrain on train + validation before test. Validation interactions inform model selection and the test seen-item mask, but are not added to the training graph or training examples.

Report **full-catalog Recall@10/20 and NDCG@10/20**, averaged equally across eligible users. "Full catalog" means all movies in the defined catalog remaining after user-specific masking. Use identical users, splits, targets, and candidate sets across configurations. Also report results by user activity and item popularity, with groups defined from train. Sampled-candidate metrics, if included, are secondary.

Global-threshold limitation and user audit. The fixed positive threshold of ≥ 4 does not account for differences in users' rating habits. Users without training positives cannot be evaluated under this setup, while users without training dislikes remain eligible through fallback.

Report:

- Realized train/validation/test sizes and timestamp boundaries.
- Eligible user counts and positives-per-user distributions for validation and test.

- Users without training positives and users without eligible held-out targets.
- Training explicit-negative counts within the chosen catalog, including users with zero, 1–2, and at least 3 such ratings.
- The share of training users requiring fallback and the realized explicit-negative sampling proportion in each mixed configuration.
- Counts of users and interactions excluded by the catalog and evaluation rules.

Statistical analysis. Run each completed configuration with at least three random seeds. Report mean performance and standard deviation across seeds.

For paired user-level comparisons, first average each user's metric across seeds within each configuration. Compute differences between configurations on the same users and obtain 95% paired bootstrap confidence intervals using 10,000 user resamples. These intervals describe uncertainty across users conditional on the fitted runs; variability across training seeds is reported separately.

Use a paired Wilcoxon signed-rank test on the seed-averaged user results as a complementary check where appropriate. Apply Holm correction to a prespecified family of comparisons. Use NDCG@20 for the primary significance comparisons:

- LightGCN versus MF at 0% and at 50% explicit negatives.
- 50% versus 0% explicit negatives within each model.

Treat the 25% configurations and other metric comparisons as sensitivity analyses unless additional confirmatory tests are specified before examining test results. Report effect sizes and intervals; significance is not guaranteed.

Main risks. Offline results depend on temporal boundaries, candidate eligibility, historical exposure, and negative sampling. Unrated does not mean disliked, and explicit low ratings are imperfect evidence. Global temporal splitting may reduce evaluation coverage, while restricting the catalog excludes movies without positive training history. Report these exclusions and interpret results within the chosen population and protocol.

Future Work

The planned extension is the 25% explicit-negative condition for both models. LightGCN++-style modifications, popularity-aware or hard-negative sampling, graph-contrastive learning, and user-relative rating thresholds are reserved for future work.

Tune both main models fairly on validation. If LightGCN does not outperform MF, investigate and report that result; architectural improvement is not required for a successful comparison.

Expected outcome. A from-scratch implementation and a controlled comparison of MF + BPR and LightGCN + BPR across three explicit-negative proportions, with a Popularity baseline, a common temporal split, fully specified candidate and masking rules, a documented coverage audit, and statistical analysis that distinguishes user-level uncertainty from variability across training runs.

Compute Resources

The team has access to GPUs through Google Colab and Kaggle and will select a suitable platform for the experiments; using both is not required. We plan to use GPU acceleration for training, hyperparameter tuning, and full-catalog evaluation. The main computational load is expected from repeated runs of different configurations with multiple random seeds. A pilot run will assess runtime and memory requirements, and checkpoints will be saved to resume work if a session ends.